Diagnosing Mitigation Need in Histopathology Patch Classification

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September 10, 2026

Abstract

Long-tailed histopathology datasets can contain many patches but few independent patients per class. This benchmark separates nominal patch shortage from patient shortage on BRACS and TCGA-UT using frozen Virchow2 features. Severe nominal deprivation damages discrimination on both datasets and is substantially recoverable with prevalence-based methods. Reducing patient coverage at fixed patch counts instead lowers balanced accuracy by 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT. Class weighting does not repair this loss, and methods that instead reshape the training objective, the sampling distribution, the decision rule, or the classifier head recover at most about one sixth of the TCGA-UT deficit. Probability-quality gains do not consistently accompany discrimination recovery; most benefits of prevalence-based weighting disappear after validation-fitted temperature scaling. The clearest unresolved target is therefore performance under limited patient support. Follow-up experiments should distinguish unequal patient influence from missing patient variation, then test patient-aware training against strong weighting and calibration baselines. Including severe class imbalance and conditions with little measured damage would establish both targeted benefit and limited harm outside the intended regime.

Contents

1 Introduction

The research goal is to develop a method that improves discrimination and/or calibration on histopathology datasets, particularly when class support is long-tailed. Patch counts alone may be insufficient for this purpose: thousands of patches from a small number of patients need not provide the variation needed to generalize to unseen patients. This report uses controlled allocation to ask which shortages cause damage, which existing interventions repair it, and where a new mechanism is justified.

The benchmark covers BRACS, a seven-class breast-lesion task, and TCGA-UT, a 30-class cancer-type task, using frozen Virchow2 features. It separates nominal patch support from independent patient support and describes the accompanying difficulty and diversity profiles. The five confirmatory methods share a class-weighted loss and differ in the signal used to set weights; a wider exploratory roster tests other mechanisms. This distinction narrows both the information a method should use and the intervention it should apply. The companion protocol specifies the design and inference (Queisler, 2026a); dataset provenance and the method taxonomy are documented separately (Queisler, 2026b,c). The protocol poses three questions, restated here so that the results can be read without the companion document at hand:

RQ1. Which shortage does controlled allocation create, and what does it damage? Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

Sections 2 and 3 answer the first two. Section 4 reports the association models for the third. Statistical tests are restricted to the five weighting methods and to comparisons meeting each dataset's prespecified materiality and precision criteria; estimates for the wider method roster and the two-group association models remain exploratory.

2 Allocation-Induced Damage

Both datasets meet the allocation, patient-support, and completeness requirements. Moderate and severe allocations reach their target ranges; concentrated and spread comparisons retain identical class patch counts while changing patient breadth. Pooled patient support passes the resampling checks. Each dataset contains three partitions, five initialization seeds per method block, and all 378 planned comparisons.

The analysis comprises 24 comparison units: three class assignments crossed with four deprived conditions on each dataset. Appendix A documents construction checks, support limits, and tuning. The following sections first describe the realized shortages, then examine their effects on discrimination, probability quality, and class tiers.

2.1 Realized signal profiles

The conditions change two things: how patches are divided among classes, and how many patients supply those patches. *Concentrated* allocations draw more patches from fewer patients;

spread allocations distribute the same class-specific patch counts across more patients. Four comparisons separate their effects:

Moderate patch shortage (moderate). Redistribute a fixed patch budget so that the largest class has about ten times as many patches as the smallest. Compare with concentrated balanced training to measure the damage from moderate class imbalance.

Severe patch shortage (severe). Increase that ratio to about one hundred, using the same concentrated balanced reference. This tests whether stronger class imbalance causes damage that moderate imbalance does not.

Severe patch shortage with broad patient coverage (severe spread). Keep severe class imbalance but spread each class's patches across patients. Compare with balanced spread training to test whether class imbalance remains harmful with broader patient coverage.

Patient shortage (independent-support deprivation). Compare concentrated balanced training with balanced spread training. Every class keeps the same patch count, but its patches come from fewer patients. This isolates the effect of replacing independent patient evidence with more patches from the remaining patients.

The short names in parentheses are used throughout the results and tables. In detailed tables, "Balanced" denotes the concentrated allocation used to induce patient shortage; "Balanced (spread)" is its reference. Severe spread and patient shortage are therefore different comparisons: the former changes class balance under broad coverage, whereas the latter changes patient coverage at equal class counts.

Table 1 describes what each allocation actually changes. The imbalance ratio ρ compares the largest and smallest class counts; $1 - H_{\text{norm}}$ measures departure from equal class shares, with zero denoting balance. The shortage scores S_{nom}, S_{ind}, and S_{div} describe nominal patch support, independent patient support, and feature diversity. Each score is standardized across the twelve comparisons within its dataset: subtract the mean shortage and divide by its population standard deviation. A negative score therefore means a smaller shortage than the dataset's average comparison, not necessarily an improvement over the balanced reference.

Pilot difficulty d_c records how hard a class is to recognize when scarcity is not the issue: it is one minus the recall a class-balanced linear probe reaches on class c in pilot draws that give every class the same number of patients and patches. It is measured once, before the conditions are built, so it describes the class rather than the allocation. The alignment A is the correlation between log class patch counts and d_c: negative A means the harder classes received fewer patches, positive A that they received more. A shortage score has to grow with harm, and the harmful case here is $A < 0$; difficulty shortage is therefore the standardized value of $-A$, and the displayed A is not itself a fourth standardized score. To identify the dominant shortage, exclude axes with a non-positive raw shortage in that comparison or no variation across the twelve comparisons, then select the largest remaining signed standardized score. If no axis remains, the comparison has no matched label and contributes only descriptively. If the two largest eligible scores differ by less than 0.25, the result is ambiguous.

Diversity is an exploratory axis. No established method in the surveyed literature reads a diversity estimate to decide when to intervene (Queisler, 2026c), so the benchmark carries one purpose-built implementation, and no condition varies diversity independently of the patch allocation: S_{div} moves with S_{nom} under the same manipulation. A targeted follow-up asked whether that coupling is a design convenience or a property of the representation. Holding each condition's realized patch counts and support profile fixed, it re-selected only *which* patches occupy the pinned slots — those nearest the slot mean, a farthest-point sample, or the allocation's original draw — and then repeated the swap with case- and slide-level pinning relaxed to an unrestricted class-level pool, the most freedom the construction can offer. The resulting swing in semantic-scale log-volume stayed one to two orders of magnitude below the roughly two-fold shortage that patch-count manipulation reaches on both datasets: at these patch counts the two extremal selections already span most of the representation's within-class covariance, leaving little volume for an identity swap to move. Diversity and patch count are therefore not merely confounded by this design but hard to separate in this representation at all, which is why no manipulated diversity condition is reported and every diversity result is read as exploratory.

Assignment	ρ	$1 - H_{\text{norm}}$	S_{nom}	S_{ind}	S_{div}	A	Dominant
BRACS — Patient shortage: concentrated balanced vs. balanced spread							
Native	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Aligned	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Reversed	1.00	0.000	-1.47	1.73	-1.31	0.20	Independent support
BRACS — Moderate patch shortage vs. concentrated balanced							
Native	10.00	0.128	-0.35	-0.58	-0.79	0.36	Nominal support
Aligned	10.00	0.128	-0.17	-0.58	-0.35	-0.92	Difficulty
Reversed	10.00	0.128	-0.39	-0.58	-0.65	0.92	Nominal support
BRACS — Severe patch shortage vs. concentrated balanced							
Native	100.38	0.353	0.76	-0.58	0.80	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	1.28	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.79	0.92	Ambiguous
BRACS — Severe spread vs. balanced spread							
Native	100.38	0.353	0.76	-0.58	1.00	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	0.88	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.99	0.92	Diversity
TCGA-UT — Patient shortage: concentrated balanced vs. balanced spread							
Native	1.00	0.000	-1.38	1.66	-1.56	0.01	Independent support
Aligned	1.00	0.000	-1.38	1.66	-1.56	0.01	Independent support
Reversed	1.00	0.000	-1.38	1.66	-1.56	0.01	Independent support
TCGA-UT — Moderate patch shortage vs. concentrated balanced							
Native	10.00	0.061	-0.34	-0.71	0.18	0.09	Diversity
Aligned	10.00	0.061	-0.14	-0.71	0.29	-0.99	Difficulty
Reversed	10.00	0.061	-0.63	-0.71	0.24	0.99	Diversity
TCGA-UT — Severe patch shortage vs. concentrated balanced							
Native	100.17	0.179	0.87	-0.71	1.14	0.09	Diversity
Aligned	90.43	0.174	1.43	-0.71	1.23	-0.99	Ambiguous
Reversed	100.17	0.179	0.32	-0.71	1.25	0.99	Diversity
TCGA-UT — Severe spread vs. balanced spread							
Native	100.17	0.179	0.87	-0.71	-0.15	0.09	Nominal support
Aligned	90.43	0.174	1.43	-0.24	0.25	-0.99	Ambiguous
Reversed	100.17	0.179	0.32	0.25	0.25	0.99	Ambiguous

Table 1: Realized shortages for the four comparisons. Shortage scores S are standardized within each dataset; negative values mean below-average shortage. A is the unstandardized support–difficulty correlation. Dominance compares eligible signed scores, including standardized $-A$, with a 0.25-standard-deviation ambiguity margin.

The independent-support comparison holds patch allocation fixed while changing patient breadth. BRACS keeps 13,982–27,202 patches fixed within each partition and expands from 78–89 concentrated patients to 103 spread patients. TCGA-UT holds 36,988–52,204 patches fixed by partition while expanding from 671–917 concentrated patients to 4,983 spread patients. The patient-shortage condition therefore removes a much larger share of the available patients on TCGA-UT than on BRACS.

Within each dataset, the three concentrated-balanced rows reuse one training allocation and differ only in the class assignment used for comparison; they are correlated views, not independent replications. The profile table therefore describes realized conditions rather than multiplying the number of independent experiments.

2.2 Discrimination deficit

The discrimination deficit D_{BA} is the balanced accuracy lost relative to the appropriate reference, reported in percentage points. The native assignment follows clinical class order; the difficulty-aligned assignment gives the fewest patches to the hardest classes, whereas the reversed assignment gives those classes the most patches. The dataset-specific minimum is $\max(1, 2\sigma_{\mathrm{seed}})$: 1.443 points on BRACS and 1.000 point on TCGA-UT. Recovery is analysed when the deficit reaches this minimum and its paired 95% interval excludes zero.

On BRACS, moderate imbalance produces deficits of 1.50–8.17 percentage points, concentrated severe imbalance 4.30–13.23, and severe spread 4.15–13.77; only aligned moderate lacks a precise material deficit. On TCGA-UT, moderate deficits are only 0.13–0.45 points and remain below threshold, whereas concentrated severe deficits reach 1.33–1.79 points and severe-spread deficits 2.40–4.96 points. Independent-support deprivation damages every assignment on both datasets, by 2.62–3.03 points on BRACS and 4.76–4.83 on TCGA-UT, despite identical patch counts within each comparison. Full estimates and intervals appear in Appendix Table 13.

TCGA-UT tolerates moderate redistribution. A tenfold head-to-tail ratio costs it 0.36, 0.13, and 0.45 percentage points under the native, aligned, and reversed assignments; the native and reversed intervals exclude zero, so the loss is real, but all three estimates stay below the one-point materiality minimum. Damage appears only at the hundredfold ratio, and even there it reaches at most 1.79 points — roughly a third of what removing patient breadth costs at unchanged patch counts (4.76–4.83). Ordering the manipulations by damage on this dataset therefore puts patient shortage first and class imbalance a distant second. Part of that tolerance is arithmetic rather than robustness: spread over 30 classes, a tenfold ratio disturbs class shares only half as much as on seven-class BRACS ($1 - H_{\mathrm{norm}}$ of 0.0609 against 0.128), so nominally matched severities are the milder manipulation here. Figure 1 shows both effects.

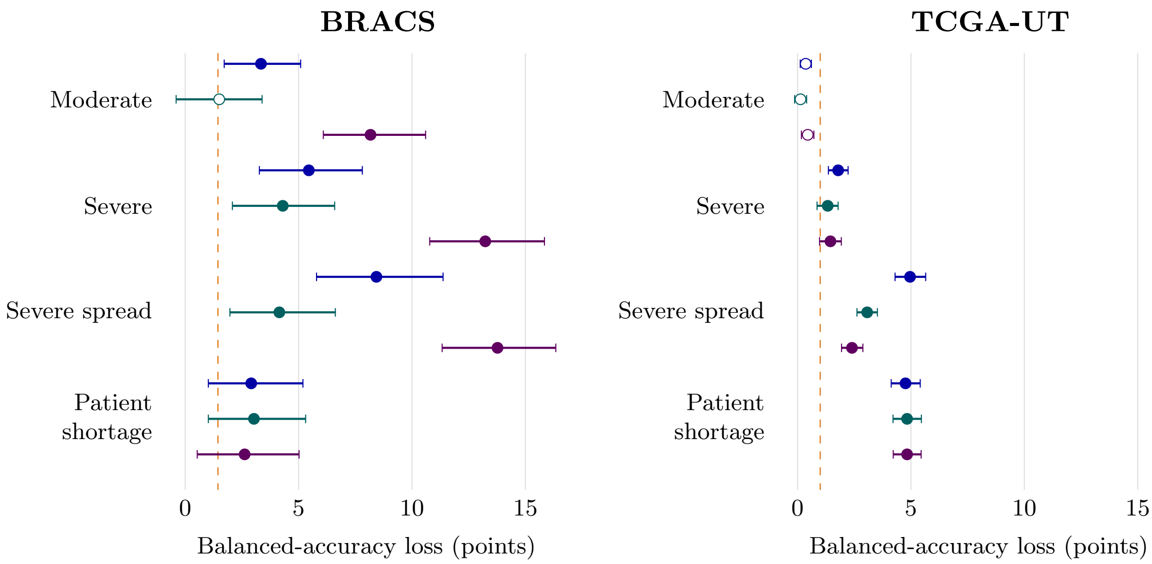


Figure 1: Patient shortage damages both datasets; moderate patch shortage has a much smaller effect on TCGA-UT. Points and bars show balanced-accuracy deficits and paired 95% intervals. Within each comparison, rows are native (blue), aligned (teal), and reversed (purple), from top to bottom. Filled points meet both materiality and precision criteria; hollow points do not. Dashed lines mark each dataset's materiality minimum. Both panels use the same horizontal scale. References are concentrated balanced for moderate/severe and balanced spread for severe spread/patient shortage.

The panels also differ in how widely the three assignments spread within one condition. On BRACS the assignment matters about as much as the severity: moderate deprivation costs between 1.50 and 8.17 points depending on which classes are starved. TCGA-UT's three assignments instead stay within roughly one point of each other in every condition, with visibly tighter intervals. Both features follow from class count rather than from a different mechanism. Balanced accuracy averages over 30 classes on TCGA-UT and seven on BRACS, so re-ordering which classes receive the scarce budget moves the aggregate by at most a thirtieth per class, and the seed noise that average must overcome is smaller (σ_{seed} of at most 0.50 points against 0.72). The manipulation is also milder at a matched imbalance ratio: $\rho \approx 100$ produces a $1 - H_{\text{norm}}$ of 0.179 on TCGA-UT against 0.353 on BRACS. Matched severities are consequently not matched doses.

2.3 Probability-quality deficit

Probability quality measures how much probability the model assigns to the correct class, using tail-group macro negative log likelihood (NLL). For readability, deficits are expressed here as the relative reduction in the geometric-mean probability assigned to the true class. If the NLL increase is D_{NLL}, this reduction is

$$Q = 100\big(1 - \exp(-D_{\text{NLL}})\big)\%. \tag{1}$$

For example, a 20% reduction means a reference geometric-mean true-class probability of 0.50 would fall to 0.40. This is a relative probability change, not a percentage-point change in accuracy or an arithmetic mean of probabilities. It retains the endpoint's patient, class, and partition weighting. Negative values indicate improvement. The thresholds derived from balanced-condition variation correspond to reductions of 13.7% on BRACS and 5.9% on TCGA-UT. The equivalent NLL thresholds are 0.14782 and 0.06095 *nats*, respectively, where a nat is a natural-log unit. Recovery eligibility still uses the prespecified NLL threshold and a paired 95% interval excluding zero.

On BRACS, moderate patch shortage reduces geometric-mean true-class probability by 17.0–53.0%, severe patch shortage by 49.7–82.5%, and severe spread by 18.5–67.7%. Native moderate is imprecise, and independent-support reductions of 12.5–13.1% fall just below threshold. TCGA-UT differs: moderate patch shortage improves this probability measure, while all three independent-support comparisons reduce it by 22.7–23.1%. Among severe concentrated conditions, aligned alone clears threshold; all three severe-spread conditions do. Probability-quality damage therefore depends on dataset and assignment and does not simply follow accuracy damage. Appendix Table 14 retains the NLL estimates and intervals.

2.4 Where the damage falls

Aggregate deficits do not show which support tier absorbs the loss. Table 2 therefore retains one informative tier for each difficulty-based assignment and dataset; the complete side-by-side tier comparison appears in Appendix D. Tier labels in balanced-spread rows reflect class ordering rather than unequal support because that allocation is uniform.

Dataset	Assignment	Tier	Balanced spread	Moderate	Severe	Severe spread
BRACS	Aligned	Tail	40.9	18.3	11.8	13.2
BRACS	Reversed	Body	31.2	19.7	15.3	13.3
TCGA-UT	Aligned	Tail	66.1	55.7	51.9	54.5
TCGA-UT	Reversed	Head	67.0	58.2	57.5	66.4

Table 2: Selected tier recall (%). Aligned assigns the fewest patches to the hardest classes; reversed assigns those classes the most patches. Head, body, and tail denote high-, intermediate-, and low-support tiers.

Against the balanced-spread anchor, BRACS shows substantial tier differences already at moderate severity: aligned tail recall falls from 40.9% to 18.3%, while reversed body recall falls from 31.2% to 19.7%. TCGA-UT moves less at moderate severity, in line with its sub-threshold aggregate deficit, but severe deprivation lowers aligned tail recall from 66.1% to 51.9%. Under the reversed TCGA-UT assignment, severe spread restores head recall to 66.4%, close to the 67.0% spread reference. These concentrated-to-spread comparisons combine patch allocation and patient coverage; they are not the matched-reference deficits of Section 2.2. The directly matched aligned severe-spread comparison still shows pronounced tail loss: 40.9% to 13.2% on BRACS and 66.1% to 54.5% on TCGA-UT. Support rank therefore needs to be interpreted alongside class difficulty and the reference allocation.

The tier pattern also explains why the reversed assignment damages BRACS more than the aligned one (13.23 against 4.30 points at severe deprivation), although giving the hardest classes the most patches looks like the gentler manipulation. Extra patches buy little recall on classes that are hard for reasons other than scarcity: against the balanced-spread anchor, reversed severe deprivation raises UDH recall only from 41.4% to 44.7% and lowers PB from 44.1% to 36.4%. The starved easy classes have far more to lose, with normal tissue falling from 67.2% to 22.2% and invasive carcinoma from 87.6% to 60.5%. The enriched borderline classes also absorb neighbouring ones: ADH reaches 55.3% recall at an F_1 of 0.173, so most of its predicted mass is wrong. Because balanced accuracy weights all classes equally, the aligned assignment removes recall where it was already low, whereas the reversed assignment removes it where it was high. TCGA-UT shows no comparable gap between assignments (1.45 against 1.33 points), consistent with a flatter and less mutually confusable set of 30 classes. Appendix D reports the class-level values.

2.5 Natural anchor

The natural condition trains on every eligible patch at each dataset's observed prevalence. It is not size-matched and enters no estimand; it only places the controlled results beside the training distribution encountered without intervention.

Dataset	Natural patches per partition	Natural BA	Concentrated balanced BA	Balanced spread BA
BRACS	193,010–217,704	49.8%	49.2%	50.4–50.9%
TCGA-UT	1,116,020–1,121,580	79.1%	75.3%	79.6–79.7%

Table 3: Natural and balanced cross-entropy anchors.

On BRACS, natural training uses roughly seven to sixteen times the controlled patch total across partitions yet reaches 49.8% balanced accuracy, between the concentrated and spread balanced references. On TCGA-UT, the natural anchor reaches 79.1%, well above the concentrated balanced result of 75.3% and close to the spread references at 79.6–79.7%. This is consistent with the controlled patient-coverage result on TCGA-UT, but the larger patch budget prevents attributing the natural anchor's performance to coverage alone.

3 Mitigation Recovery

Recovery divides improvement over deprived cross-entropy by the established deficit. For balanced accuracy, the numerator is BA(method) − BA(deprived CE); for NLL, it is NLL(deprived CE) − NLL(method), since lower NLL is better. Both are reported as percentages. A recovery of 0% means no improvement over deprived cross-entropy, 100% reaches the balanced reference, a negative value means further deterioration, and a value above 100%

surpasses the reference. Comparisons that do not meet the materiality and precision criteria in Sections 2.2 and 2.3 are omitted because their recovery denominator is not sufficiently stable.

3.1 Which weighting signals recover discrimination

The confirmatory family contains five weighted cross-entropy methods that differ only in the signal used to set their class weights: prevalence, nominal support, independent support, pilot difficulty, or semantic diversity. Each is compared with deprived cross-entropy using a paired permutation test. A result *survives Holm adjustment* when its adjusted p-value remains below 0.05 after accounting for all confirmatory comparisons, reducing the chance that a result appears significant only because many methods and conditions were tested.

Dataset	Shortage	Units	Balanced-accuracy recovery	Holm result
BRACS	Independent	3	All signals: −6.5–2.5%.	0/15
BRACS	Nominal	8	Prevalence/nominal: 42.6–89.2%.	10/16 improve
TCGA-UT	Independent	3	All signals: −1.9–0.0%.	0/15
TCGA-UT	Severe	3	Prevalence/nominal: 41.9–85.3%.	6/6 improve
TCGA-UT	Severe spread	3	Prevalence/nominal: 32.1–80.7%; independent: 8.1–47.7%.	8 improve; 1 deteriorates

Table 4: Balanced-accuracy recovery by shortage and weighting signal. Ranges are percentages of the original deficit; the final column counts results that remain significant after Holm adjustment.

Prevalence and nominal-support weighting are the only signals that repeatedly repair nominal deprivation. On BRACS they recover 42.6–89.2% across the eight eligible nominal units, with ten of sixteen comparisons surviving Holm adjustment. TCGA-UT's moderate units do not enter recovery analysis because their deficits are below threshold; across its six severe units, the same two signals recover 32.1–85.3% and all six concentrated-severe comparisons survive adjustment. Severe-spread recovery is more mixed: independent-support weighting also recovers 8.1–47.7%, and difficulty weighting produces one adjusted deterioration. No weighting repairs independent-support deprivation on either dataset: all estimates lie between −6.5% and 2.5% on BRACS and between −1.9% and 0.0% on TCGA-UT. Appendix Table 15 provides the condition-level estimates and intervals.

Two construction details bound what the difficulty and diversity estimates can mean. Pilot difficulty is frozen from a balanced pilot before the conditions exist, so its weights are identical in every condition and assignment and cannot track the allocation that caused the deficit; because $d_c \in [0, 1]$, its weight ratio stays near a factor of two, whereas prevalence and nominal-support weights scale with $\rho = 100$. Semantic-scale weighting is estimated on draws with equal cases and patches per class, a deliberate choice that keeps the estimate from encoding prevalence, so it has no direct reading of a nominal shortage. Consistent with this, tuning selected the weakest available difficulty exponent in 23 of 26 selections, against the strongest available prevalence and nominal-support setting in 18 and 22; since the difficulty grid does not contain $\tau = 0$, that selection is left-censored. These two arms therefore test two specific weight maps rather than the difficulty and diversity signals themselves.

Figure 2 retains each assignment: the failure to repair patient shortage is shared across signals and datasets, whereas nominal-deprivation recovery varies substantially.

3.2 Matched against unmatched signals

A method is *matched* when its weighting signal corresponds to the dominant shortage identified for that condition in Table 1; for example, independent-support weighting is matched to a condition that reduces patient breadth. The unmatched arm contains the other weighting signals. Conditions without one clear dominant shortage are excluded from this contrast. A

BRACS	Prevalence	Nominal	Independent	Difficulty	Diversity
Moderate: native	82*	89*	20	12	14
Moderate: aligned	—	—	—	—	—
Moderate: reversed	84*	83*	5	-8	-6
Severe: native	77	64	11	-27	-17
Severe: aligned	65	43	2	16	-6
Severe: reversed	79*	74*	11	-6	-1
Severe spread: native	87*	83*	11	2	-3
Severe spread: aligned	47	46	-14	-14	-14
Severe spread: reversed	82*	79*	5	-1	3
Patient shortage: native	0	1	1	-6	2
Patient shortage: aligned	0	0	1	-6	2
Patient shortage: reversed	0	1	2	-7	3

TCGA-UT	Prevalence	Nominal	Independent	Difficulty	Diversity
Moderate: native	—	—	—	—	—
Moderate: aligned	—	—	—	—	—
Moderate: reversed	—	—	—	—	—
Severe: native	76*	72*	-1	-5	13
Severe: aligned	42*	43*	0	0	-6
Severe: reversed	85*	81*	1	-12	-5
Severe spread: native	74*	73*	48*	-2	2
Severe spread: aligned	34*	32*	8	3	2
Severe spread: reversed	80*	81*	25*	-13*	-8
Patient shortage: native	0	0	-1	-2	0
Patient shortage: aligned	0	0	-1	-2	0
Patient shortage: reversed	0	0	-1	-2	0

Figure 2: Weighting repairs nominal deprivation more readily than patient shortage. Cells show balanced-accuracy recovery (%), rounded to whole percentages: 0 means no repair and 100 reaches the balanced reference. Teal indicates improvement and orange deterioration; colour intensity saturates at an absolute recovery of 100%, while numbers remain untruncated. Asterisks mark Holm-adjusted $p < 0.05$ from the permutation tests; paired intervals appear in Appendix C. Grey dashes mean the original deficit is ineligible, not zero recovery. Columns denote the five weighting signals, with nominal and independent referring to support.

positive matched-minus-unmatched value therefore favours the signal selected by the condition profile.

Dataset	Dominant shortage	Units	Matched minus unmatched balanced accuracy (percentage points)	Holm result
BRACS	Independent support	3	0.06 [−0.13, 0.22]	No difference
TCGA-UT	Independent support	3	0.00 [−0.11, 0.12]	No difference
BRACS	Nominal support	2	2.34 [0.96, 3.75]; 7.08 [5.25, 9.15]	Both favour matched
TCGA-UT	Nominal support	1	2.86 [2.36, 3.34]	Favours matched
Both	Difficulty or diversity	6	Matched weighting never outperforms effective prevalence-based alternatives.	Two favour unmatched

Table 5: Matched-minus-unmatched contrasts for balanced accuracy. Intervals are paired 95% bootstrap intervals.

For independent-support deprivation, matched weighting changes balanced accuracy by only 0.06 percentage points on BRACS and 0.00 on TCGA-UT; both intervals cover zero. Matching favours the corresponding signal in the two BRACS nominal-support units and the one TCGA-UT nominal-support unit. When difficulty or diversity is dominant, the matched method never outperforms the effective prevalence-based alternatives and is worse in two units. Those units are patch-shortage conditions: dominance is assigned from standardized scores, and no condition creates a difficulty or diversity shortage while holding patch counts fixed. The contrast therefore compares weighting rules within a nominal deprivation.

Both independent-support and class-balanced weighting use an effective-number formula whose parameter β controls weight strength. The matched-β diagnostic gives independent-support weighting the value selected for class-balanced cross-entropy, separating the count source from tuning strength. It does not alter the conclusion on either dataset. Full matched contrasts and matched-β results appear in Appendix Tables 17 and 19.

3.3 Recovery of probability quality

A method can restore balanced accuracy while making its probabilities worse. The prespecified recovery endpoint is tail-group macro negative log likelihood computed from raw test probabilities; temperature scaling never enters the confirmatory decisions. Recovery is analysed only where the dataset-specific raw-score deficit is sufficiently large and precise.

Dataset	Condition group	Units	Raw tail-NLL recovery pattern
BRACS	Nominally deprived	8	Prevalence and nominal-support weighting improve seven units by 53.9–122.7%; both are negative in native severe spread. Other signals deteriorate in all 24 comparisons.
TCGA-UT	Independent support	3	Every weighting signal deteriorates by 51.7–62.4%; all 15 comparisons survive Holm adjustment.
TCGA-UT	Aligned severe	1	Prevalence and nominal-support weighting recover 106.5% and 92.2%; other signals deteriorate.
TCGA-UT	Severe spread	3	Native improves, aligned is mixed, and reversed deteriorates under four of five signals; the response depends on assignment.

Table 6: Recovery of tail-group probability quality. Recovery is the percentage of the original negative-log-likelihood deficit.

On BRACS, prevalence and nominal-support weighting improve raw probability quality in seven of eight eligible units, whereas the other three signals deteriorate in all 24 comparisons. TCGA-UT does not repeat that simple pattern. All five signals worsen the three independent-support units by 51.7–62.4% of the original deficit, while prevalence and nominal-support weighting recover 106.5% and 92.2% in aligned concentrated severe. Across severe-spread assignments the result changes from improvement to mixed response to deterioration. Probability-quality recovery is therefore conditional on dataset and assignment even when discrimination recovery is consistent. Exact intervals and adjusted p-values appear in Appendix Table 20; Appendix F gives the full temperature-scaling analysis summarized below.

Temperature scaling fits a single confidence adjustment on validation data and leaves predicted classes unchanged. Repeating the tail-group contrasts after scaling removes much of the nominal-deprivation effect: among the raw-eligible nominal comparisons, three of eight BRACS deficits and two of four TCGA-UT deficits still exceed their dataset's magnitude threshold. Some nominal deficits change sign, including aligned severe on TCGA-UT. Patient shortage behaves differently. TCGA-UT's geometric-mean true-class probability remains 20.4–20.7% below its spread reference after scaling, compared with 22.7–23.1% before scaling; all three paired intervals still exclude zero. On BRACS, the corresponding deficit decreases from 12.5–13.1% before scaling to 9.7–10.3% after scaling. Scaling therefore reduces the independent-support deficit but does not eliminate its positive point estimate. Both ranges are below BRACS's 13.7% materiality threshold; being below that threshold does not mean that the deficit is zero. Nor does scaling eliminate all BRACS probability-quality deficits: reversed severe, reversed severe spread, and aligned severe spread retain deficits above the magnitude threshold.

The apparent probability-quality benefit of prevalence and nominal-support weighting largely disappears after scaling. On BRACS, the number of improvements with intervals excluding zero falls from fourteen of sixteen comparisons to one, and four estimates change to deterioration. On TCGA-UT, all fourteen comparisons for these two methods become deteriorations with intervals excluding zero. Independent-support, difficulty, and diversity weighting also retain deterioration in every evaluated comparison, although scaling reduces its magnitude. Thus TCGA-UT's scaled method effects have a consistent direction across assignments: none of the five weighting signals improves probability quality relative to scaled deprived cross-entropy. This directional consistency concerns mitigation effects; it does not mean every allocation-induced deficit has the same sign. Scaled contrasts are exploratory paired-bootstrap results, without the confirmatory permutation tests or multiplicity adjustment used for the raw endpoint.

3.4 Exploratory methods beyond weighting

The wider roster varies the training objective, sampling distribution, decision rule, or classifier head. Its estimates are exploratory because methods no longer differ by signal alone; several also reduce to the same balanced-sampling mechanism and should not be read as independent confirmations.

The roster confirms that severe nominal-support deficits are broadly repairable: the best estimates are 55.4–90.8% on BRACS and 90.0–130.6% on TCGA-UT. Independent-support damage remains largely unrecovered. No BRACS method has positive recovery with an interval excluding zero. TCGA-UT's CFAL and OKO estimates are positive but close only 12.9–16.6% of the deficit, while classifier retraining deteriorates by about 13.5%. Thus no tested method closes a substantial share of the independent-support loss.

The roster also separates signal from mechanism on the nominal axis. Focal loss and CFAL read example difficulty and recover 38–108% of the severe concentrated deficits, against −27–16% for difficulty weighting on the same units. Both couple difficulty with prevalence or nominal support, so this does not isolate difficulty; it does show that difficulty-sensitive mechanisms repair deficits a static class-difficulty weight leaves in place. No roster method reads a diversity

Dataset	Endpoint and shortage	Units	Most informative result
BRACS	Balanced accuracy, nominal support	8	Best method recovers 79.2–90.8% in seven units; 55.4% in aligned severe spread.
BRACS	Balanced accuracy, independent support	3	No positive recovery with an interval excluding zero; OKO, focal loss, and cRT worsen performance.
TCGA-UT	Balanced accuracy, severe shortage	6	Several mechanisms recover most of the deficit; best estimates range from 90.0% to 130.6%.
TCGA-UT	Balanced accuracy, independent support	3	CFAL and OKO recover only 12.9–16.6%; cRT deteriorates by 13.4–13.5%.

Table 7: Most informative recovery results from the exploratory method roster.

signal, so diversity has no second reading. Full method-level estimates appear in Appendix Table 22.

4 Explaining Heterogeneity

Damage magnitude and recovery headroom vary across conditions, and RQ3 asks which realized shortages track that variation. Two association models are specified, one for signed balanced-accuracy damage over the cross-entropy cells and one for recovery over the gate-passing method cells. Both use the same linear predictor over standardized quantities,

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\text{ind}} S_{\text{ind},j} + \beta_A A_j + \beta_{\text{div}} S_{\text{div},j}, \tag{2}$$

where ρ_j is achieved severity, $S_{\text{ind},j}$ the independent-support shortage, A_j the support–difficulty alignment, $S_{\text{div},j}$ the diversity shortage, and $u_{g[j]}$ a random intercept over dataset–target groups that absorbs level differences between datasets. Observation errors come from the bootstrap distributions of the corresponding endpoints.

The fitted sample consists of BRACS and TCGA-UT. For each model, predictors are centered and divided by their standard deviations across the observations entering that fit. These model scales differ from the within-dataset shortage scores used to select matched methods in Table 1. Each coefficient describes the change associated with a one-standard-deviation increase in its predictor, holding the other predictors fixed.

Model	α	$\beta_{\log\rho}$	β_{ind}	β_A	β_{div}	σ_u	σ
Damage	0.0416	0.0318	0.0158	0.0100	-0.0163	0.1436	0.0208
Recovery	0.2652	-0.0427	-0.3385	0.0631	0.0045	0.1418	0.4541

Table 8: Maximum a posteriori estimates for the two exploratory association models fitted to BRACS and TCGA-UT. Damage is signed balanced-accuracy loss and recovery is the recovered fraction. Predictors are standardized within each model's fitted sample.

Conditional on the other profile variables, achieved severity and independent-support shortage have positive damage coefficients, 0.0318 and 0.0158. In the recovery model, independent-support shortage has the largest coefficient in magnitude and is negative, -0.3385; severity is also negative, while alignment and diversity are small and positive. The damage model's diversity coefficient is negative, reversing the expected direction; the model conditions on achieved severity rather than nominal shortage while S_{div} moves with the patch allocation, so it should not be read as a protective effect of low diversity. These are exploratory associations, not causal effects, and two dataset–target groups cannot establish that the coefficients transport to a new dataset.

Leave-one-dataset-out prediction is poor: errors reach 6.18 balanced-accuracy percentage points on BRACS and 13.96 on TCGA-UT (Appendix E). The model supplies hypotheses for controlled follow-up, not a transferable rule for selecting mitigation.

5 Discussion: From Diagnosis to Method Design

The benchmark narrows the design problem to two complementary aims: recover discrimination when independent patient support is limited, and improve probability quality without sacrificing discrimination. Severe nominal deprivation already has effective baselines. Patient shortage remains largely unrepaired, while the calibration analysis shows why accuracy recovery alone is an insufficient success criterion.

5.1 What the results rule in and out

Prevalence and nominal-support weighting provide strong baselines for severe patch shortage (Section 3.1). A further per-class weight would need to improve on these baselines, rather than demonstrate recovery against cross-entropy alone. Conversely, weak difficulty and diversity weights do not establish that those signals are unhelpful. Their construction limits their response to allocation, and the wider roster changes several mechanisms at once. Its difficulty-sensitive results justify testing example-level interventions, but do not identify difficulty as the source of their advantage.

Independent patient support is the stronger unresolved target. Its controlled reduction damages discrimination on both datasets at unchanged class patch counts, and even the wider roster leaves most of that deficit intact. However, this does not yet identify a trainable remedy. Unequal patch contributions may let particular patients dominate optimization, or the available patients may omit variation needed at test time. Patient-aware weighting can address the first explanation; it cannot supply missing observations in the second. The benchmark establishes damage from reduced coverage, not that dependence-aware training will recover it.

The link to long-tailed learning also needs a direct test. The cleanest patient-shortage comparison uses balanced class counts. It establishes that independent support matters beyond nominal imbalance, but does not establish how a patient-aware intervention behaves when rare classes have both few patches and few patients. Severe-spread comparisons show that broader coverage alone does not remove class-allocation damage. A useful method may therefore need separate controls for class balance and patient influence.

5.2 Discrimination and calibration require separate decisions

NLL evaluates probability quality, which depends on both discrimination and calibration; it is not a pure calibration measure. Temperature scaling provides a limited diagnostic by adjusting confidence while preserving predicted classes. On TCGA-UT, the five weighting signals all deteriorate relative to scaled deprived cross-entropy in the evaluated tail-group comparisons. On BRACS, most prevalence-based benefits also disappear after scaling (Section 3.3). A raw NLL gain alone is consequently weak evidence for a contribution beyond standard confidence adjustment.

Patient shortage nevertheless leaves a positive NLL deficit after scaling on both datasets, exceeding the magnitude threshold on TCGA-UT but not BRACS. This residual motivates further work, but does not isolate class-specific miscalibration from impaired discrimination. Moreover, the benchmark selected methods for balanced accuracy. Its NLL results describe those operating points, not the limits of probability-focused selection.

A follow-up should declare whether its primary aim is discrimination, probability quality, or a constrained trade-off between them. Every baseline and candidate should receive the same opportunity for validation-fitted calibration. Tail-group NLL should be accompanied by class- or tier-level reliability diagnostics and recall, so that confidence changes can be distinguished from changed class decisions. Aggregate ECE alone would be insufficient evidence of tail repair.

5.3 A staged experiment to identify a useful mechanism

The next experiments should resolve the two patient-shortage explanations before introducing a more complex model. Table 9 orders the proposed comparisons. These are prospective hypotheses and decision criteria, not additional benchmark findings.

Experiment	Controlled comparison	Decision it supports
Patient influence	At fixed patients and class patch counts, vary concentration of patch contributions; compare patch-average and patient-average training.	A benefit that grows with concentration supports unequal patient influence as a repairable mechanism.
Patient breadth	At fixed class patch counts, vary contributing patients and repeat patient selection.	Residual loss after equalizing patient influence identifies a limitation that simple reweighting does not resolve.
Long-tailed transfer	Cross patient breadth with balanced, moderate, and severe class allocation; ablate class balancing and patient averaging separately and together.	Benefit beyond class balancing under joint shortage supports a method for the intended regime.
Probability quality	Compare raw and validation-calibrated predictions under accuracy-focused and NLL-focused selection.	A gain over calibrated baselines within a prespecified accuracy tolerance supports a probability-quality contribution.

Table 9: Follow-up experiments from mechanism diagnosis to long-tailed evaluation. Match patch budgets, feature extractor, classifier capacity, and training exposure within each comparison.

The first candidate is a patient-average loss: average patch losses within a patient, then give patients equal influence. For the long-tailed comparison, compute patient averages within each class before combining classes, so that patient influence and class balance can be varied separately. This matters when one patient contributes to several classes. Compare ordinary cross-entropy, the effective prevalence or nominal-support baseline, patient averaging alone, and their combination. Equal patient sampling with within-patient patch sampling can estimate the same objective; it is an implementation control, not an independent mechanism. When patients contribute equal patch counts within a class, patient and patch averaging coincide there, providing a check on the proposed explanation.

Vary patient breadth and concentration of contributions as separately as feasibility allows. Repeat patient-subset draws because changing breadth also changes which patients are observed; initialization seeds alone do not measure that variation. Hold class-specific patch totals fixed within each breadth comparison, report contributions per patient, and quantify feature redundancy separately. Equal patient weights do not necessarily remove redundant examples. If the patch inventory prevents a fully crossed design, report feasible contrasts rather than interpreting coupled changes as independent effects.

Advance the simple objective only if it improves on class balancing under joint nominal and patient shortage. If it helps only with highly unequal contributions, its scope is correction of patient dominance. If equalizing influence leaves the coverage deficit unchanged, adding patient diversity is the immediate data-level remedy; a model-based extension would need information beyond scalar weights. A later experiment could test patient-aware regularization or sharing across related classes, with a matched-capacity ablation of the sharing mechanism. The present evidence does not justify choosing a particular architecture.

The probability-quality experiment can use the same training candidates. Compare accuracy-focused selection with validation NLL selection subject to a prespecified balanced-accuracy tolerance. Separate calibration fitting from calibrator selection using validation partitions or cross-fitting, keeping test patients untouched. If validation patients support a persistent class-specific error, compare a regularized class-specific calibrator with global temperature scaling. Limited tail support is a reason to constrain complexity, not evidence that a more flexible calibrator will help.

5.4 What would count as progress

Success requires a material paired improvement over a strong baseline, not only positive recovery against deprived cross-entropy. A discrimination-focused method should improve balanced accuracy within a prespecified probability-quality tolerance; a probability-focused method should improve tail NLL beyond calibrated baselines within an accuracy tolerance. Report absolute endpoint differences, uncertainty, and tier recall alongside recovery. Retain balanced and moderate controls even where the original deficit is below threshold: harm remains measurable although a recovery ratio is unstable.

Include a zero-strength setting and choose intervention strength on patient-disjoint validation data. The paired test-set deficits in this benchmark identify promising experimental regimes, but cannot serve as a deployment activation rule: the balanced counterfactual is unavailable and test labels must not guide intervention. Any adaptive rule must use training support and validation evidence, then be frozen before evaluation.

The current data can motivate candidates and ablations. Final confirmation requires fresh held-out patients or another dataset, with hypotheses, endpoint priorities, meaningful effect sizes, and multiplicity handling fixed in advance. Preserve patient dependence across repeated partitions. Test joint shortages and difficulty assignments explicitly rather than counting the three balanced patient-shortage views as independent replications. Match processed examples and tuning budgets, and report additional memory and runtime; the roster's cost differences make simple weighting an essential practical comparator (Appendix H).

RQ3 should not determine an adaptive rule. Its leave-one-dataset-out errors are large relative to observed damage (Appendix E); the associations motivate investigation of patient support but cannot predict intervention strength on a new dataset. Likewise, BRACS's greater moderate-allocation sensitivity could reflect class-share disruption, baseline difficulty, patient support, or feature suitability. Matching support and class-share disruption across tasks would help separate these explanations; the present comparison cannot.

6 Scope and Conclusion

These findings concern patch classification with frozen Virchow2 features. They do not establish the same limitations for representation learning or whole-slide prediction, where labelled-slide allocation couples nominal and independent support. The three partitions per dataset are repeated partitions, not independent cohorts. Within each dataset, the three patient-shortage comparisons reuse one deprived allocation and differ in reference assignment. Patient-shortage doses differ across datasets, so their damage magnitudes do not isolate dataset susceptibility. Only nominal and independent support are manipulated; difficulty and diversity conclusions remain conditional on their realized profiles and tested implementations.

Severe nominal imbalance has effective existing remedies, whereas reduced patient coverage leaves a replicated discrimination deficit that those remedies scarcely recover. Probability quality remains a separate target, with calibration-adjusted baselines necessary to establish added value. The next step is to test whether controlling patient influence recovers part of the coverage deficit under long-tailed allocation. That experiment will determine whether a simple patient-aware objective is sufficient or whether progress requires additional patient variation or a mechanism that shares information across limited observations.

A What Was Held Fixed, and What Was Checked

The supporting analyses address four questions: whether the intended allocations were achieved, whether the estimates have adequate patient support, whether recovery depends on the assignment, and whether calibration or partition variation changes the interpretation. Repeated execution records are summarized as ranges; paired uncertainty intervals are retained for the primary damage and weighting comparisons. Ranges describe observed variation, not confidence intervals.

A.1 Allocation and patient coverage

Table 10 separates class imbalance from patient coverage. Within each partition, balanced and balanced-spread training use identical class-specific patch counts. Their patient counts differ substantially, especially on TCGA-UT. Moderate and severe allocations meet their target ratios of 9–11 and 90–110, respectively; none is off-target or collapses to balance. Small departures from a ratio of one in balanced training reflect integer allocation.

Allocation	Patients	Patches	ρ	$1 - H_{\mathrm{norm}}$
BRACS				
Natural	103	193,010 to 217,704	5.722 to 8.297	0.1178 to 0.1432
Balanced	78 to 89	13,982 to 27,202	1.000 to 1.001	0.0000
Balanced spread	103	13,982 to 27,202	1.000 to 1.001	0.0000
Moderate	78 to 89	13,982 to 27,202	10.003 to 10.007	0.1280
Severe	78 to 89	13,982 to 27,202	100.301 to 100.467	0.3530
Severe spread	103	13,982 to 27,202	100.301 to 100.467	0.3530
TCGA-UT				
Natural	4,983	1,116,020 to 1,121,580	17.850 to 18.395	0.0754 to 0.0765
Balanced	671 to 917	36,988 to 52,204	1.001	0.0000
Balanced spread	4,983	36,988 to 52,204	1.001	0.0000
Moderate	671 to 917	36,988 to 52,204	9.992 to 10.003	0.0609
Severe	671 to 917	36,988 to 52,204	90.107 to 100.500	0.1735 to 0.1792
Severe spread	4,203 to 4,983	36,988 to 52,204	90.107 to 100.500	0.1735 to 0.1792

Table 10: Allocation ranges across the three partitions and available assignments. Patients are unique contributing patients, not summed class counts. Patch counts refer to one realized training allocation. Spread increases patient coverage at matched patch counts; it does not necessarily reach every eligible patient under severe allocation.

For BRACS, balanced training uses 13,982–27,202 patches from 78–89 patients, compared with 103 patients under balanced spread. The corresponding TCGA-UT comparison uses 36,988–52,204 patches from 671–917 versus 4,983 patients. The patient inventory is therefore exhausted by balanced spread on both datasets, while TCGA-UT severe spread reaches 4,203–4,983 patients.

A.2 Patient support and complete comparisons

The *bootstrap patient-support check* evaluates the planned resampling before training, using only patient identifiers, partition membership, and class labels. It checks whether each pooled class has at least five contributing patients, whether the 2.5th percentile of its Kish effective count is at least five, and whether a single patient carries more than half the weight in at most 5% of replicates. Failure would restrict the dataset to descriptive estimates.

Both datasets pass. The closest effective-count margins are 5.5 for ductal carcinoma in situ on BRACS and 6.6 for uterine carcinosarcoma on TCGA-UT. No pooled class has a dominant patient in any replicate.

The qualification changes at the individual partition level. BRACS flags atypical ductal hyperplasia, ductal carcinoma in situ, and flat epithelial atypia. TCGA-UT flags adrenocortical carcinoma, kidney chromophobe, mesothelioma, ovarian serous cystadenocarcinoma, pheochromocytoma and paraganglioma, rectum adenocarcinoma, thymoma, uterine carcinosarcoma, and uveal melanoma. Class-specific estimates from those partitions need caution; the confirmatory comparisons use the valid pooled patient support.

All five confirmation seeds and all three partitions are present in every required block, yielding 378 planned comparisons per dataset. Relative achieved-severity spread is at most 0.002 on BRACS and 0.010 on TCGA-UT, below the 0.5 limit. Semantic-scale weighting records zero invalid draws across 195 reweighting pools per dataset: its fallback to unit weights never occurs.

A.3 What tuning selected

Training budgets count example presentations, not optimizer updates: controlled and natural budgets are 503,880 and 6,521,970 on BRACS, and 1,343,760 and 33,480,600 on TCGA-UT. A method consuming e_m examples per update receives $\lceil E/e_m \rceil$ updates, with checkpoint spacing adjusted to keep approximately the same number of validation passes.

Tables 11 and 12 show which parameter values were selected and how often the strength was zero. They summarize the 392 recorded condition–assignment–method entries, including configurations reused across assignments; these are not 392 independent tuning exercises. No entry selects a learning rate at the boundary of the candidate range. This rules out observed boundary selections, but does not establish that a wider range could not improve validation performance.

Method	Learning rate	Strength	Zero strength
Balanced sampling	0.00001 to 0.00010	0.2500 to 1.0000	0/13
CE	0.00003 to 0.00300	—	—
CE + soft F_1	0.00003 to 0.00010	0.0000	13/13
CE + soft MCC	0.00001 to 0.00030	0.0000 to 64.0000	9/13
CFAL	0.00010 to 0.00300	0.2500 to 1.0000	0/13
Nominal support	0.00003 to 0.00030	0.9990 to 0.9999	0/13
cRT	0.00300 to 0.01000	—	—
Focal loss	0.00003 to 0.00030	0.0000 to 1.0000	6/13
Independent support	0.00003 to 0.00100	0.8000 to 0.9990	0/13
Logit adjustment (train)	0.00003 to 0.00030	0.2500 to 2.0000	0/13
OKO	0.00030 to 0.00300	1.0000 to 4.0000	0/13
Difficulty	0.00003 to 0.00100	0.2500 to 0.5000	0/13
Logit adjustment (post-hoc)	Reused CE	0.5000 to 1.0000	0/13
Diversity	0.00003 to 0.00300	0.2500 to 1.0000	0/13
Prevalence	0.00003 to 0.00010	0.7500 to 1.0000	0/13

Table 11: BRACS tuning summary. Ranges span recorded selections; zero-strength counts use entries with a strength parameter as denominator. Strength has a method-specific meaning and cannot be compared numerically across methods.

Method	Learning rate	Strength	Zero strength
Balanced sampling	0.00003 to 0.00010	0.0000 to 1.0000	3/13
CE	0.00010 to 0.00030	—	—
CE + soft F_1	0.00003 to 0.00010	1.0000 to 16.0000	0/13
CE + soft MCC	0.00003 to 0.00010	1.0000 to 16.0000	0/13
CFAL	0.00010 to 0.00030	0.2500 to 4.0000	0/13
Nominal support	0.00003 to 0.00010	0.9900 to 0.9999	0/13
cRT	0.00100 to 0.00300	—	—
Focal loss	0.00003 to 0.00010	0.0000	13/13
Independent support	0.00010 to 0.00030	0.8000 to 0.9990	0/13
Logit adjustment (train)	0.00010	0.5000 to 2.0000	0/13
OKO	0.00030 to 0.00100	1.0000 to 4.0000	0/13
Difficulty	0.00010 to 0.00030	0.2500	0/13
Logit adjustment (post-hoc)	Reused CE	0.2500 to 1.0000	0/13
Diversity	0.00010 to 0.00030	0.2500 to 1.0000	0/13
Prevalence	0.00003 to 0.00010	0.0000 to 1.0000	3/13

Table 12: TCGA-UT tuning summary, using the same conventions as Table 11. Post-hoc logit adjustment reuses the cross-entropy checkpoint.

Zero strength does not always mean ordinary cross-entropy. The soft-F_1 and soft-MCC hybrids retain balanced sampling; focal loss at $\gamma = 0$ retains class-aware α. These selected candidates were trained with those mechanisms intact. Most methods receive 16 candidates, cross-entropy and classifier retraining four, and post-hoc logit adjustment one shard without new training.

Selection maximizes natural-prevalence validation balanced accuracy, with macro F_1 and negative log likelihood as successive tie-breakers. It pools assignments and does not optimize the tail-group probability-quality endpoint. Consequently, weak recovery can reflect both the signal and the selected operating point; a calibration deterioration describes a method tuned for discrimination.

The defining estimator is patient-macro, equivalent to the protocol's case-macro estimator for BRACS and TCGA-UT. It pools each patient's partition appearances, matching the crossed patient bootstrap.

B How Large Was the Damage?

The natural condition provides context but is not the reference for these deficits. Moderate and severe concentrated training are compared with concentrated balanced training; severe spread and patient shortage use balanced spread. Positive values mean worse performance under deprivation. Tables 13 and 14 preserve every assignment and its paired interval, making the distinction between an imprecise deficit and a sub-threshold one explicit.

Dataset	Comparison	Assignment	Deficit [95% CI]	Eligible
BRACS	Patient shortage	Native	2.91 [1.02, 5.19]	Yes
BRACS	Patient shortage	Aligned	3.03 [1.02, 5.31]	Yes
BRACS	Patient shortage	Reversed	2.62 [0.53, 5.02]	Yes
BRACS	Moderate	Native	3.34 [1.72, 5.09]	Yes
BRACS	Moderate	Aligned	1.50 [-0.40, 3.39]	Imprecise
BRACS	Moderate	Reversed	8.17 [6.09, 10.60]	Yes
BRACS	Severe	Native	5.45 [3.27, 7.81]	Yes
BRACS	Severe	Aligned	4.30 [2.08, 6.59]	Yes
BRACS	Severe	Reversed	13.23 [10.78, 15.84]	Yes
BRACS	Severe spread	Native	8.43 [5.79, 11.37]	Yes
BRACS	Severe spread	Aligned	4.15 [1.97, 6.62]	Yes
BRACS	Severe spread	Reversed	13.77 [11.33, 16.34]	Yes
TCGA-UT	Patient shortage	Native	4.76 [4.13, 5.41]	Yes
TCGA-UT	Patient shortage	Aligned	4.83 [4.21, 5.46]	Yes
TCGA-UT	Patient shortage	Reversed	4.83 [4.22, 5.45]	Yes
TCGA-UT	Moderate	Native	0.36 [0.12, 0.61]	Below minimum
TCGA-UT	Moderate	Aligned	0.13 [-0.13, 0.40]	Below minimum
TCGA-UT	Moderate	Reversed	0.45 [0.17, 0.72]	Below minimum
TCGA-UT	Severe	Native	1.79 [1.36, 2.23]	Yes
TCGA-UT	Severe	Aligned	1.33 [0.86, 1.79]	Yes
TCGA-UT	Severe	Reversed	1.45 [0.97, 1.93]	Yes
TCGA-UT	Severe spread	Native	4.96 [4.30, 5.65]	Yes
TCGA-UT	Severe spread	Aligned	3.07 [2.62, 3.52]	Yes
TCGA-UT	Severe spread	Reversed	2.40 [1.94, 2.88]	Yes

Table 13: Balanced-accuracy loss in percentage points, with paired 95% intervals. Eligibility requires both the dataset's minimum deficit and an interval excluding zero. TCGA-UT moderate has small positive estimates, not an established absence of loss.

Dataset	Comparison	Assignment	Deficit [95% CI]	Eligible
BRACS	Patient shortage	Native	0.138 [0.064, 0.216]	Below minimum
BRACS	Patient shortage	Aligned	0.140 [0.071, 0.213]	Below minimum
BRACS	Patient shortage	Reversed	0.134 [0.061, 0.209]	Below minimum
BRACS	Moderate	Native	0.186 [-0.007, 0.397]	Imprecise
BRACS	Moderate	Aligned	0.379 [0.157, 0.715]	Yes
BRACS	Moderate	Reversed	0.754 [0.180, 1.706]	Yes
BRACS	Severe	Native	0.688 [0.332, 1.099]	Yes
BRACS	Severe	Aligned	0.729 [0.488, 1.045]	Yes
BRACS	Severe	Reversed	1.744 [1.208, 2.295]	Yes
BRACS	Severe spread	Native	0.205 [0.100, 0.334]	Yes
BRACS	Severe spread	Aligned	0.908 [0.575, 1.267]	Yes
BRACS	Severe spread	Reversed	1.129 [0.830, 1.474]	Yes
TCGA-UT	Patient shortage	Native	0.263 [0.226, 0.304]	Yes
TCGA-UT	Patient shortage	Aligned	0.259 [0.224, 0.296]	Yes
TCGA-UT	Patient shortage	Reversed	0.258 [0.224, 0.295]	Yes
TCGA-UT	Moderate	Native	-0.199 [-0.244, -0.152]	Below minimum
TCGA-UT	Moderate	Aligned	-0.118 [-0.153, -0.084]	Below minimum
TCGA-UT	Moderate	Reversed	-0.062 [-0.093, -0.038]	Below minimum
TCGA-UT	Severe	Native	-0.168 [-0.226, -0.100]	Below minimum
TCGA-UT	Severe	Aligned	0.143 [0.081, 0.219]	Yes
TCGA-UT	Severe	Reversed	-0.031 [-0.076, 0.020]	Below minimum
TCGA-UT	Severe spread	Native	0.193 [0.139, 0.263]	Yes
TCGA-UT	Severe spread	Aligned	0.199 [0.151, 0.256]	Yes
TCGA-UT	Severe spread	Reversed	0.081 [0.042, 0.121]	Yes

Table 14: Tail-group negative-log-likelihood increase in nats, with paired 95% intervals. Negative values mean improved probability quality. TCGA-UT moderate improves this endpoint under all three assignments, even though balanced accuracy falls slightly.

This contrast matters for interpretation: moderate deprivation on TCGA-UT is not simply too noisy to detect. Native and reversed balanced-accuracy intervals exclude zero, but their estimates remain below the one-point minimum. All three probability-quality intervals instead favour moderate training. Neither finding makes the much larger natural training set an appropriate causal reference.

C Which Signals Recover the Loss?

The main-text recovery map shows the pattern; the following tables retain the uncertainty behind it. Each cell gives recovery as a percentage, its paired bootstrap 95% interval, and the Holm-adjusted permutation p-value. Zero means no repair and 100% means the balanced reference is reached. Negative recovery means further damage. Only eligible comparisons appear; absent rows must not be interpreted as zero recovery. Nominal and independent denote the corresponding support-weighted cross-entropy methods.

Comparison	Prevalence	Nominal	Independent	Difficulty	Diversity
Moderate Native	81.5 [41.9, 123.7] $p = 0.0130$	89.2 [54.9, 130.7] $p = 0.0022$	19.7 [-19.0, 55.7] $p = 1.0000$	12.1 [-26.3, 45.3] $p = 1.0000$	14.2 [-41.5, 62.2] $p = 1.0000$
Moderate Reversed	84.0 [72.0, 96.2] $p = 0.0026$	83.3 [69.6, 96.9] $p = 0.0011$	5.1 [-14.4, 20.2] $p = 1.0000$	-7.8 [-24.4, 5.9] $p = 1.0000$	-6.4 [-27.9, 12.6] $p = 1.0000$
Severe Native	76.8 [44.1, 103.0] $p = 0.1214$	63.7 [35.3, 88.0] $p = 0.1297$	10.6 [-14.9, 32.3] $p = 1.0000$	-27.3 [-63.7, -7.0] $p = 0.0603$	-16.5 [-54.8, 4.9] $p = 1.0000$
Severe Aligned	65.4 [33.0, 106.1] $p = 0.3344$	42.6 [0.2, 76.7] $p = 1.0000$	2.4 [-19.3, 21.4] $p = 1.0000$	16.4 [-7.1, 43.7] $p = 1.0000$	-5.9 [-52.4, 23.7] $p = 1.0000$
Severe Reversed	79.4 [69.1, 88.9] $p = 0.0006$	73.6 [61.9, 85.6] $p = 0.0006$	11.1 [4.6, 18.4] $p = 0.0505$	-5.9 [-12.0, 0.2] $p = 0.8122$	-1.0 [-9.2, 6.4] $p = 1.0000$
Severe spread Native	87.1 [70.9, 105.8] $p = 0.0016$	83.1 [67.5, 101.3] $p = 0.0006$	11.1 [-3.3, 22.7] $p = 1.0000$	2.1 [-17.3, 19.0] $p = 1.0000$	-2.6 [-15.1, 7.4] $p = 1.0000$
Severe spread Aligned	46.6 [0.5, 82.3] $p = 1.0000$	45.5 [8.5, 75.5] $p = 1.0000$	-13.5 [-46.1, 5.9] $p = 1.0000$	-14.5 [-83.6, 20.4] $p = 1.0000$	-14.5 [-61.6, 11.2] $p = 1.0000$
Severe spread Reversed	82.4 [71.4, 94.0] $p = 0.0006$	79.1 [68.5, 90.6] $p = 0.0006$	5.1 [-0.6, 11.3] $p = 0.8772$	-0.8 [-9.1, 5.9] $p = 1.0000$	3.0 [-3.7, 9.2] $p = 1.0000$
Patient shortage Native	0.3 [-1.4, 3.1] $p = 1.0000$	0.5 [-1.3, 3.3] $p = 1.0000$	1.3 [-0.8, 5.5] $p = 1.0000$	-5.8 [-25.6, 6.6] $p = 1.0000$	2.2 [-23.0, 30.0] $p = 1.0000$
Patient shortage Aligned	0.3 [-1.4, 3.0] $p = 1.0000$	0.4 [-1.2, 3.2] $p = 1.0000$	1.3 [-0.8, 5.2] $p = 1.0000$	-5.6 [-24.9, 6.2] $p = 1.0000$	2.1 [-22.3, 28.1] $p = 1.0000$
Patient shortage Reversed	0.3 [-2.1, 4.0] $p = 1.0000$	0.5 [-1.8, 4.3] $p = 1.0000$	1.5 [-1.0, 8.2] $p = 1.0000$	-6.5 [-37.2, 8.1] $p = 1.0000$	2.5 [-34.1, 34.7] $p = 1.0000$

Table 15: BRACS discrimination recovery (%). Prevalence and nominal-support weighting repair nominal deprivation, while patient shortage remains unrepaired. Each cell retains estimate, interval, and adjusted p-value.

Comparison	Prevalence	Nominal	Independent	Difficulty	Diversity
Severe Native	76.4 [57.9, 95.3] $p = 0.0005$	71.9 [54.3, 89.5] $p = 0.0005$	-1.0 [-12.5, 10.0] $p = 1.0000$	-5.3 [-16.4, 6.4] $p = 1.0000$	12.6 [-2.0, 25.6] $p = 1.0000$
Severe Aligned	41.9 [20.6, 63.0] $p = 0.0115$	43.0 [22.8, 65.2] $p = 0.0063$	-0.2 [-15.3, 9.4] $p = 1.0000$	0.2 [-11.7, 11.8] $p = 1.0000$	-6.3 [-29.9, 10.2] $p = 1.0000$
Severe Reversed	85.3 [70.3, 99.7] $p = 0.0005$	81.2 [65.4, 95.7] $p = 0.0005$	1.2 [-9.5, 10.0] $p = 1.0000$	-11.6 [-31.0, 2.4] $p = 0.4280$	-4.8 [-30.9, 15.1] $p = 1.0000$
Severe spread Native	73.7 [65.5, 81.2] $p = 0.0005$	73.3 [65.4, 80.7] $p = 0.0005$	47.7 [39.1, 55.8] $p = 0.0005$	-1.9 [-7.2, 2.8] $p = 1.0000$	2.0 [-4.8, 8.2] $p = 1.0000$
Severe spread Aligned	34.1 [25.0, 43.2] $p = 0.0005$	32.1 [21.9, 42.3] $p = 0.0005$	8.1 [-2.3, 17.7] $p = 1.0000$	2.8 [-5.5, 9.9] $p = 1.0000$	2.1 [-7.6, 11.6] $p = 1.0000$
Severe spread Reversed	79.6 [69.7, 90.2] $p = 0.0005$	80.7 [71.5, 90.8] $p = 0.0005$	24.6 [14.4, 35.0] $p = 0.0101$	-13.0 [-23.6, -3.8] $p = 0.0061$	-8.0 [-18.9, 2.4] $p = 1.0000$
Patient shortage Native	-0.2 [-0.9, 0.5] $p = 1.0000$	0.0 [-0.1, 0.1] $p = 1.0000$	-0.5 [-3.3, 2.0] $p = 1.0000$	-1.9 [-3.3, -0.4] $p = 0.6938$	-0.3 [-4.9, 3.7] $p = 1.0000$
Patient shortage Aligned	-0.2 [-0.9, 0.5] $p = 1.0000$	0.0 [-0.1, 0.1] $p = 1.0000$	-0.5 [-3.2, 2.0] $p = 1.0000$	-1.8 [-3.2, -0.4] $p = 0.6938$	-0.3 [-4.7, 3.7] $p = 1.0000$
Patient shortage Reversed	-0.2 [-0.9, 0.5] $p = 1.0000$	0.0 [-0.1, 0.1] $p = 1.0000$	-0.5 [-3.2, 1.9] $p = 1.0000$	-1.8 [-3.2, -0.4] $p = 0.6938$	-0.3 [-4.7, 3.7] $p = 1.0000$

Table 16: TCGA-UT discrimination recovery (%). Moderate is omitted because it fails the materiality criterion. Independent-support weighting helps severe spread in some assignments but does not repair patient shortage itself.

A useful distinction is between improving a deprived model and outperforming a different weighting signal. The matched-minus-unmatched contrast tests the latter, and excludes ambiguous shortage profiles. Tables 17 and 18 retain the eligible, unambiguous contrasts; positive values favour the matched signal. Their units are endpoint differences, not recovery percentages.

Dataset	Comparison	Assignment	Matched signal	Contrast [95% CI]	Holm p
BRACS	Patient shortage	Native	Independent support	0.06 [-0.13, 0.22]	1.0000
BRACS	Patient shortage	Aligned	Independent support	0.06 [-0.13, 0.22]	1.0000
BRACS	Patient shortage	Reversed	Independent support	0.06 [-0.13, 0.22]	1.0000
BRACS	Moderate	Native	Nominal support	2.34 [0.96, 3.75]	0.0367
BRACS	Moderate	Reversed	Nominal support	7.08 [5.25, 9.15]	0.0006
BRACS	Severe	Aligned	Difficulty	-0.42 [-1.63, 0.86]	1.0000
BRACS	Severe spread	Aligned	Difficulty	-1.27 [-2.42, 0.05]	0.4554
BRACS	Severe spread	Reversed	Diversity	-5.29 [-6.48, -4.18]	0.0006
TCGA-UT	Patient shortage	Native	Independent support	0.00 [-0.11, 0.12]	1.0000
TCGA-UT	Patient shortage	Aligned	Independent support	0.00 [-0.11, 0.12]	1.0000
TCGA-UT	Patient shortage	Reversed	Independent support	0.00 [-0.11, 0.12]	1.0000
TCGA-UT	Severe	Native	Diversity	-0.41 [-0.68, -0.14]	0.1598
TCGA-UT	Severe	Reversed	Diversity	-0.64 [-0.91, -0.37]	0.0005
TCGA-UT	Severe spread	Native	Nominal support	2.86 [2.36, 3.34]	0.0005

Table 17: Matched-minus-unmatched balanced accuracy in percentage points. Intervals are paired bootstrap 95% intervals; p-values use the confirmatory Holm adjustment.

Dataset	Comparison	Assignment	Matched signal	Contrast [95% CI]	Holm p
BRACS	Moderate	Aligned	Difficulty	-0.557 [-1.462, 0.071]	0.0004
BRACS	Moderate	Reversed	Nominal support	1.334 [1.077, 1.595]	0.0004
BRACS	Severe	Aligned	Difficulty	-0.486 [-0.883, -0.123]	0.0004
BRACS	Severe spread	Aligned	Difficulty	-0.323 [-0.578, -0.104]	0.0004
BRACS	Severe spread	Reversed	Diversity	-0.764 [-1.032, -0.536]	0.0004
TCGA-UT	Patient shortage	Native	Independent support	0.005 [-0.003, 0.014]	0.0102
TCGA-UT	Patient shortage	Aligned	Independent support	0.005 [-0.003, 0.014]	1.0000
TCGA-UT	Patient shortage	Reversed	Independent support	0.005 [-0.003, 0.014]	1.0000
TCGA-UT	Severe spread	Native	Nominal support	0.024 [-0.007, 0.060]	0.0004

Table 18: Matched-minus-unmatched probability quality in nats, oriented so positive favours the matched signal. Ambiguous profiles are excluded.

Equalizing the effective-number parameter β tests whether independent-support weighting merely selected an unsuitable strength. Table 19 shows that this does not rescue patient shortage: BRACS balanced accuracy falls from 49.2% under cross-entropy to 48.4% with matched β; TCGA-UT remains at 75.3%. These are descriptive absolute accuracies, not paired damage estimates.

Dataset	Comparison	Assignment	CE	Independent	Matched β	Nominal
BRACS	Patient shortage	Native	49.2	49.3	48.4	49.2
BRACS	Patient shortage	Aligned	49.2	49.3	48.4	49.2
BRACS	Patient shortage	Reversed	49.2	49.3	48.4	49.2
BRACS	Moderate	Native	46.7	47.9	47.6	49.5
BRACS	Moderate	Aligned	46.0	45.6	45.7	47.7
BRACS	Moderate	Reversed	42.3	43.4	43.1	48.6
BRACS	Severe	Native	44.7	45.6	45.4	47.9
BRACS	Severe	Aligned	42.9	43.0	43.3	46.1
BRACS	Severe	Reversed	38.2	39.8	38.5	47.3
BRACS	Severe spread	Native	43.2	44.7	45.1	50.3
BRACS	Severe spread	Aligned	43.1	42.7	42.5	47.8
BRACS	Severe spread	Reversed	37.5	37.8	37.9	49.1
TCGA-UT	Patient shortage	Native	75.3	75.3	75.3	75.3
TCGA-UT	Patient shortage	Aligned	75.3	75.3	75.3	75.3
TCGA-UT	Patient shortage	Reversed	75.3	75.3	75.3	75.3
TCGA-UT	Moderate	Native	74.9	74.8	74.8	75.3
TCGA-UT	Moderate	Aligned	75.1	75.1	75.1	75.3
TCGA-UT	Moderate	Reversed	74.8	74.9	74.9	75.4
TCGA-UT	Severe	Native	73.5	73.4	73.5	74.8
TCGA-UT	Severe	Aligned	73.9	73.9	73.8	74.5
TCGA-UT	Severe	Reversed	73.8	73.9	73.9	75.1
TCGA-UT	Severe spread	Native	74.6	77.0	77.2	78.4
TCGA-UT	Severe spread	Aligned	76.7	76.9	77.0	77.7
TCGA-UT	Severe spread	Reversed	77.4	78.0	78.0	79.3

Table 19: Balanced accuracy (%) under cross-entropy, tuned independent-support weighting, independent-support weighting at nominal-support's selected β, and nominal-support weighting.

Probability-quality recovery tells a different story from discrimination. Large negative percentages occur when further NLL deterioration is divided by a comparatively small eligible deficit. They quantify damage relative to that deficit, not negative probabilities. The following two tables retain this scale rather than truncating large deteriorations.

Comparison	Prevalence	Nominal	Independent	Difficulty	Diversity
Moderate Aligned	119.7 [99.9, 153.7] $p = 0.0004$	122.7 [100.6, 170.8] $p = 0.0004$	-83.8 [-275.9, -10.5] $p = 0.0013$	-135.2 [-646.7, 39.5] $p = 0.0004$	-112.5 [-257.4, -48.1] $p = 0.0004$
Moderate Reversed	88.7 [44.1, 99.7] $p = 0.0004$	87.3 [35.3, 97.7] $p = 0.0004$	-27.4 [-442.5, 49.4] $p = 0.0098$	-109.5 [-653.3, -1.4] $p = 0.0004$	-130.3 [-830.8, 18.6] $p = 0.0004$
Severe Native	91.1 [74.3, 100.8] $p = 0.0004$	68.2 [17.7, 88.5] $p = 0.0070$	-128.7 [-321.5, -21.2] $p = 0.0015$	-216.8 [-584.7, -87.9] $p = 0.0004$	-175.3 [-484.5, -65.5] $p = 0.0004$
Severe Aligned	113.7 [104.3, 126.1] $p = 0.0004$	53.9 [6.9, 91.3] $p = 0.0004$	-250.9 [-384.0, -158.7] $p = 0.0004$	-176.8 [-320.3, -77.9] $p = 0.0004$	-357.0 [-596.8, -233.4] $p = 0.0004$
Severe Reversed	94.8 [89.6, 98.4] $p = 0.0004$	88.0 [80.7, 93.5] $p = 0.0004$	-38.8 [-85.5, -13.7] $p = 0.0013$	-115.8 [-181.2, -68.5] $p = 0.0004$	-106.0 [-195.1, -52.3] $p = 0.0004$
Severe spread Native	-40.7 [-193.8, 21.7] $p = 0.4841$	-44.2 [-201.9, 20.0] $p = 0.5092$	-323.3 [-775.3, -145.6] $p = 0.0004$	-405.8 [-888.4, -209.4] $p = 0.0004$	-389.3 [-955.2, -165.5] $p = 0.0004$
Severe spread Aligned	74.0 [57.6, 83.2] $p = 0.0004$	74.2 [58.2, 82.7] $p = 0.0004$	-54.2 [-112.5, -16.8] $p = 0.0004$	-34.5 [-71.7, -4.2] $p = 0.0004$	-89.7 [-187.8, -19.2] $p = 0.0004$
Severe spread Reversed	71.9 [59.7, 80.6] $p = 0.0004$	66.3 [48.6, 77.6] $p = 0.0004$	-111.0 [-195.0, -60.3] $p = 0.0004$	-161.2 [-255.5, -99.1] $p = 0.0004$	-101.2 [-189.0, -48.0] $p = 0.0004$

Table 20: BRACS raw tail-group NLL recovery (%), with paired intervals and Holm-adjusted p-values. Prevalence-based repair does not extend to the other three signals.

Comparison	Prevalence	Nominal	Independent	Difficulty	Diversity
Severe Aligned	106.5 [43.3, 174.7] $p = 0.0004$	92.2 [35.7, 158.8] $p = 0.0004$	-151.7 [-283.1, -91.6] $p = 0.0004$	-158.3 [-334.6, -70.8] $p = 0.0006$	-204.9 [-412.9, -118.4] $p = 0.0004$
Severe spread Native	42.8 [22.7, 59.3] $p = 0.0004$	50.6 [31.1, 67.0] $p = 0.0004$	66.4 [52.4, 75.8] $p = 0.0004$	15.5 [-32.3, 43.5] $p = 0.0004$	20.9 [0.4, 36.4] $p = 0.0004$
Severe spread Aligned	-0.7 [-34.5, 23.4] $p = 0.0004$	-0.4 [-39.3, 25.2] $p = 0.0004$	-118.9 [-196.0, -66.5] $p = 0.0264$	-58.6 [-128.5, -13.5] $p = 1.0000$	-110.1 [-185.1, -57.8] $p = 0.0004$
Severe spread Reversed	-54.3 [-180.4, -4.2] $p = 0.0004$	-53.1 [-177.7, -4.6] $p = 0.0004$	-181.3 [-433.5, -90.4] $p = 0.1713$	-177.8 [-463.4, -82.7] $p = 1.0000$	-146.8 [-391.0, -63.2] $p = 1.0000$
Patient shortage Native	-60.7 [-70.4, -52.9] $p = 0.0004$	-61.2 [-70.8, -53.5] $p = 0.0004$	-56.4 [-69.5, -45.0] $p = 0.0004$	-59.3 [-69.2, -51.2] $p = 0.0004$	-51.7 [-66.8, -37.2] $p = 0.0004$
Patient shortage Aligned	-61.7 [-71.0, -54.2] $p = 0.0287$	-62.3 [-71.6, -54.7] $p = 0.0289$	-57.4 [-69.9, -46.4] $p = 0.0102$	-60.4 [-69.9, -52.6] $p = 0.0295$	-52.6 [-67.5, -38.2] $p = 0.0010$
Patient shortage Reversed	-61.9 [-71.2, -54.3] $p = 0.0004$	-62.4 [-71.6, -54.8] $p = 0.0004$	-57.5 [-70.0, -46.5] $p = 0.0004$	-60.5 [-70.1, -52.7] $p = 0.0004$	-52.7 [-67.3, -38.4] $p = 0.0004$

Table 21: TCGA-UT raw tail-group NLL recovery (%), using the same conventions. Patient-shortage rows are eligible here, unlike on BRACS.

C.1 Does changing the mechanism help?

The wider roster is exploratory. Tables 22 and 23 summarize every roster method by its range across eligible assignments. These ranges reveal whether a method's apparent success is consistent across assignments; they are not confidence intervals or additional significance tests. The tables focus on discrimination because the confirmatory NLL tables and calibration analysis already establish that discrimination repair cannot be read as probability repair.

Method	Moderate	Severe	Severe spread	Patient shortage
Balanced sampling	72 to 80	73 to 80	44 to 91	−23 to −20
CE + soft F_1	71 to 80	67 to 79	44 to 91	−4 to −3
CE + soft MCC	71 to 80	73 to 80	44 to 91	−56 to −48
CFAL	31 to 50	38 to 59	17 to 49	−46 to −40
cRT	19 to 38	5 to 44	−26 to 43	−38 to −33
Focal loss	77 to 83	60 to 81	38 to 86	−75 to −64
Logit adjustment (train)	80 to 87	22 to 72	12 to 84	−1
OKO	46 to 74	52 to 79	55 to 83	−106 to −92
Logit adjustment (post-hoc)	23	1 to 39	−39 to 62	0

Table 22: Exploratory BRACS discrimination recovery (%): minimum to maximum across eligible assignments. A single rounded value can represent several similar estimates.

Method	Moderate	Severe	Severe spread	Patient shortage
Balanced sampling	—	54 to 99	38 to 70	4
CE + soft F_1	—	63 to 101	41 to 72	5
CE + soft MCC	—	67 to 106	41 to 74	6
CFAL	—	90 to 108	28 to 59	16 to 17
cRT	—	−10 to 13	22 to 47	−14 to −13
Focal loss	—	39 to 89	34 to 79	0
Logit adjustment (train)	—	38 to 60	32 to 91	−1
OKO	—	90 to 131	28 to 68	13
Logit adjustment (post-hoc)	—	9 to 36	16 to 72	0

Table 23: Exploratory TCGA-UT discrimination recovery (%). CFAL and OKO partly repair patient shortage, but remain far from closing the deficit. Dashes indicate no eligible comparison.

D Which Class Tiers Lose Recall?

Aggregate accuracy can hide compensation between tiers: better performance on well-supported classes may accompany worse performance on the tail. The tables below retain all three assignments and all tiers, with conditions placed side by side. They report absolute recall rather than causal differences; balanced spread is the reference only for severe spread among the nominal conditions shown. In balanced-spread rows, tier names describe class ordering rather than unequal patch counts.

Assignment	Tier	Balanced spread	Moderate	Severe	Severe spread
Native	head	51.4	49.2	49.3	51.7
Native	body	44.3	44.3	41.7	37.6
Native	tail	52.5	45.0	41.0	36.5
Aligned	head	67.0	76.4	74.7	73.8
Aligned	body	32.5	37.6	41.0	40.8
Aligned	tail	40.9	18.3	11.8	13.2
Reversed	head	40.4	46.1	48.5	51.2
Reversed	body	31.2	19.7	15.3	13.3
Reversed	tail	66.9	45.9	35.5	31.9

Table 24: BRACS tier recall (%). Aligned tail recall falls sharply even at moderate allocation, while head recall rises. The aggregate therefore combines losses and gains.

Assignment	Tier	Balanced spread	Moderate	Severe	Severe spread
Native	head	82.2	78.4	78.9	86.2
Native	body	75.3	70.8	69.6	75.0
Native	tail	81.5	75.5	71.9	62.7
Aligned	head	91.1	90.5	91.0	92.6
Aligned	body	81.9	79.2	78.8	83.0
Aligned	tail	66.1	55.7	51.9	54.5
Reversed	head	67.0	58.2	57.5	66.4
Reversed	body	82.7	77.9	77.2	79.5
Reversed	tail	89.5	88.2	86.9	86.5

Table 25: TCGA-UT tier recall (%). Moderate changes are smaller than on BRACS, but severe deprivation still damages the aligned tail.

Under aligned TCGA-UT allocation, tail recall declines from 66.1% under balanced spread to 51.9% under severe concentrated training; severe spread raises it to 54.5%. Under the reversed assignment, the head contains the difficult classes and recovers to 66.4% under severe spread. Tier identity must therefore be read together with the assignment. Individual class estimates require additional caution where the partition-level patient-support qualifications in Appendix A.2 apply.

E Can the Associations Predict Another Dataset?

Leave-one-dataset-out prediction tests a stronger claim than describing the observed comparisons. The damage model is fitted to one dataset's twelve comparison units and predicts the other's twelve; centering and scaling use the training dataset, and no unseen-dataset intercept is supplied. Training on BRACS gives a TCGA-UT root-mean-square error of 13.96 balanced-accuracy percentage points. The reverse direction gives 6.18 points. Both errors are large relative to the observed deficits. The association model therefore describes these data but does not reliably predict damage in another dataset.

F What Confidence Scaling Repairs

Temperature scaling adjusts confidence without changing the predicted class. Table 26 isolates its effect on the cross-entropy anchors: aggregate NLL and expected calibration error (ECE) before and after fitting temperature on natural-prevalence validation data. The confirmatory endpoint is tail-group NLL, so improved aggregate calibration does not itself establish tail repair.

Dataset	Assignment	Condition	Raw NLL	Scaled NLL	Raw ECE	Scaled ECE
BRACS	Aligned	Balanced spread	0.917	0.919	0.041	0.047
BRACS	Aligned	Moderate	1.211	1.047	0.143	0.059
BRACS	Aligned	Severe	1.551	1.098	0.219	0.065
BRACS	Aligned	Severe spread	1.564	1.064	0.209	0.064
BRACS	Reversed	Balanced spread	0.920	0.924	0.041	0.046
BRACS	Reversed	Moderate	1.769	1.164	0.215	0.062
BRACS	Reversed	Severe	2.582	1.271	0.313	0.067
BRACS	Reversed	Severe spread	1.915	1.180	0.259	0.048
BRACS	Native	Balanced spread	0.924	0.922	0.050	0.048
BRACS	Native	Moderate	1.274	1.111	0.148	0.077
BRACS	Native	Severe	1.793	1.199	0.220	0.079
BRACS	Native	Severe spread	1.203	1.062	0.147	0.062
BRACS	—	Patient shortage	0.993	0.995	0.051	0.055
BRACS	—	Natural	0.925	0.896	0.069	0.042
TCGA-UT	Aligned	Balanced spread	0.604	0.589	0.035	0.010
TCGA-UT	Aligned	Moderate	0.817	0.780	0.058	0.011
TCGA-UT	Aligned	Severe	1.055	0.848	0.119	0.012
TCGA-UT	Aligned	Severe spread	0.811	0.684	0.087	0.011
TCGA-UT	Reversed	Balanced spread	0.600	0.587	0.034	0.010
TCGA-UT	Reversed	Moderate	0.875	0.797	0.079	0.012
TCGA-UT	Reversed	Severe	1.191	0.881	0.140	0.010

Table 26: Cross-entropy aggregate NLL and ECE before and after temperature scaling. These are descriptive point estimates; confirmatory raw tail-group comparisons retain their paired uncertainty in Appendix C.

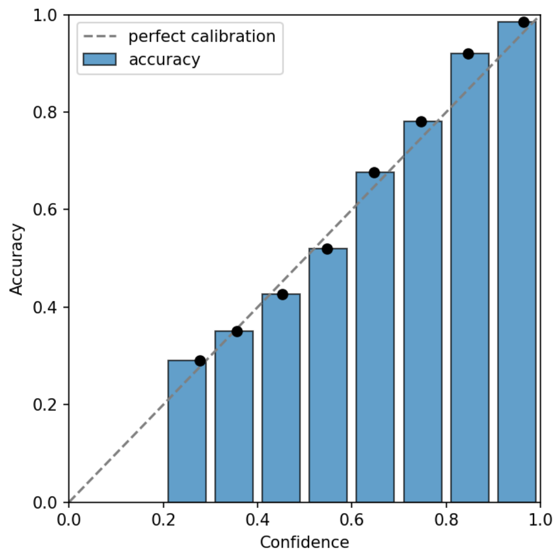
Temperature scaling reduces aggregate miscalibration on both datasets but by different amounts. For difficulty-reversed severe, BRACS expected calibration error falls from 0.313 to 0.067 and negative log likelihood from 2.582 to 1.271; on TCGA-UT the same quantities fall from 0.140 to 0.010 and from 1.191 to 0.881. This shows that a large aggregate raw-score difference can reflect correctable confidence scaling, but aggregate repair does not establish repair of the tail-group endpoint.

The aggregate metrics do not substitute for the tail-group endpoint. The contrasts of Section 3.3 are therefore re-estimated here on scaled probabilities, holding everything else fixed: the same confirmation runs, the same tail classes taken from each deprived condition's own allocation, the same frozen crossed patient bootstrap, and the same equal-weight average over the three partitions. Recomputing the raw contrasts by the same route reproduces the published values in Tables 14 and 20 exactly, which is what licenses reading the scaled column beside them. These scaled contrasts carry paired bootstrap intervals only. The partition-block permutation tests and the Holm adjustment were run on the prespecified raw endpoint and are not repeated here, so no scaled result is a confirmatory decision; the materiality minimum and the interval criterion are applied to them descriptively.

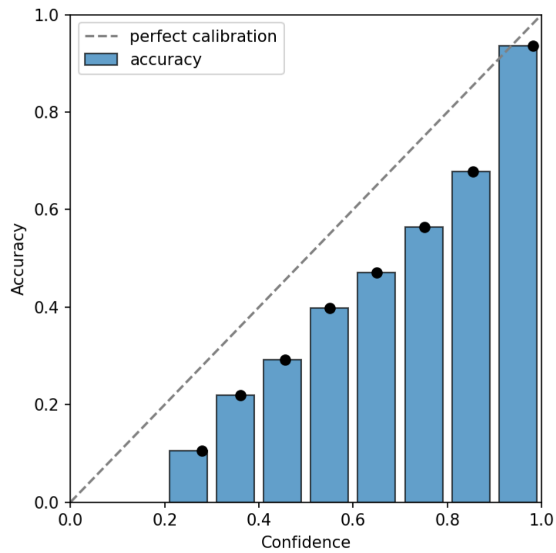
Three findings follow, and they do not point the same way. First, much of the nominal-allocation deficit is removed by adjusting confidence scale. Across BRACS's eight eligible deprived conditions the scaled deficits span -0.189 to 0.280 nats; only difficulty-reversed severe, difficulty-reversed severe spread, and difficulty-aligned severe spread still clear the 0.148-nat minimum, and difficulty-aligned moderate, difficulty-aligned severe, and native severe change sign, meaning deprived cross-entropy has lower tail-group NLL than its own balanced reference. TCGA-UT's four eligible nominal conditions behave the same way on a smaller scale: they span -0.056 to 0.106 nats, difficulty-aligned severe changes sign, and only the difficulty-aligned and native severe-spread deficits clear its 0.061-nat minimum.

Second, a single confidence adjustment does not remove the independent-support deficit, and this is where the two datasets separate. TCGA-UT's three independent-support deficits fall only from 0.258–0.263 to 0.228–0.232 nats; all three remain above the minimum with intervals excluding zero, so scaling leaves that damage essentially intact. BRACS's equivalents fall from 0.134–0.140 to 0.102–0.109 nats. These positive deficit estimates are reduced, not eliminated, and remain below its 0.14782-nat minimum both before and after scaling. The threshold qualifies the size of the deficit; it does not establish its absence. Holding patch counts fixed while removing patients therefore produces a probability-quality loss that post-hoc calibration does not repair, at the dose where the endpoint is precise enough to read.

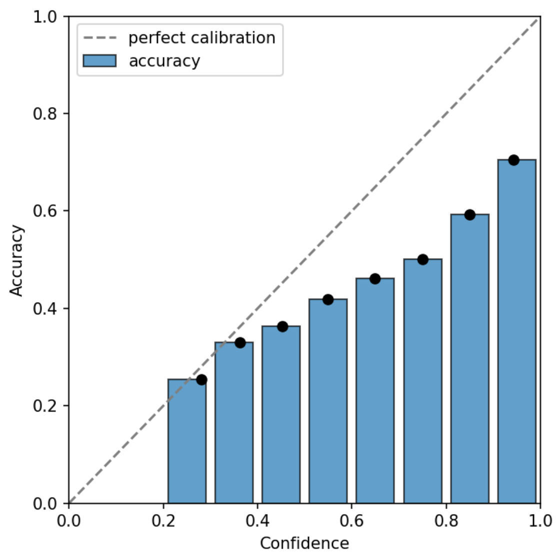
Third, the repair credited to prevalence-based weighting does not survive scaling on either dataset, whereas the harm from the other three signals does. On BRACS, 14 of the 16 raw prevalence-based improvements have intervals excluding zero against one after scaling, and four cells turn into deteriorations. On TCGA-UT the reversal is complete: four raw improvements become none, and all 14 scaled cells are deteriorations with intervals excluding zero. The remaining three signals deteriorate throughout: 21 of BRACS's 24 raw deteriorations qualify and 21 of 24 scaled ones do, with the largest compressed from 2.604 to 0.501 nats, while on TCGA-UT 18 of 21 qualify in both, compressed from 0.293 to 0.177 nats. Post-hoc calibration therefore undoes much of what the nominal allocation did, and most of what prevalence weighting was credited with repairing, but neither the harm the remaining signals cause nor TCGA-UT's independent-support damage.



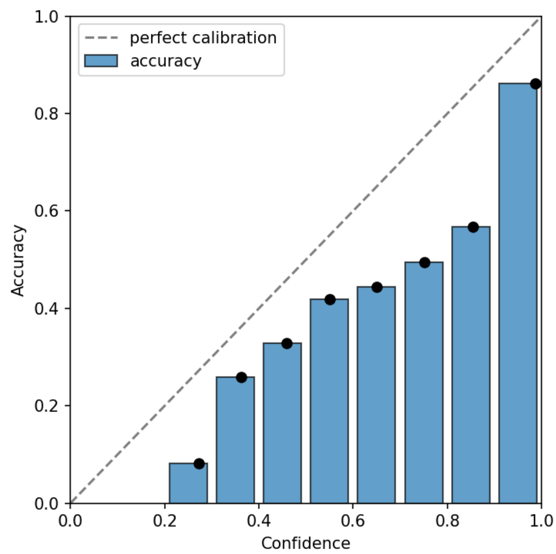
(a) Balanced reference, all classes



(b) Severe, native assignment, tail classes

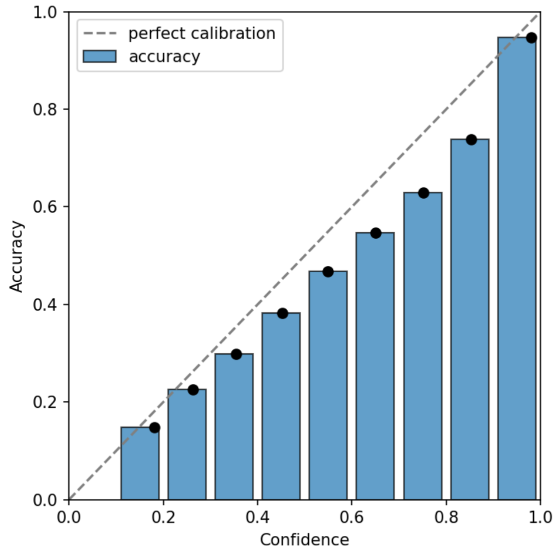


(c) Severe, aligned assignment, tail classes

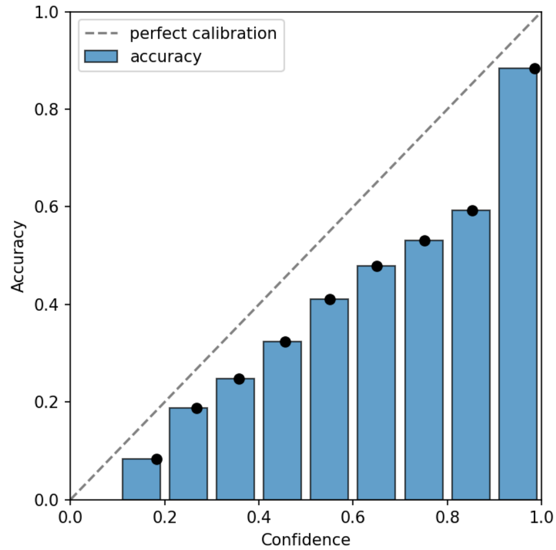


(d) Severe, reversed assignment, tail classes

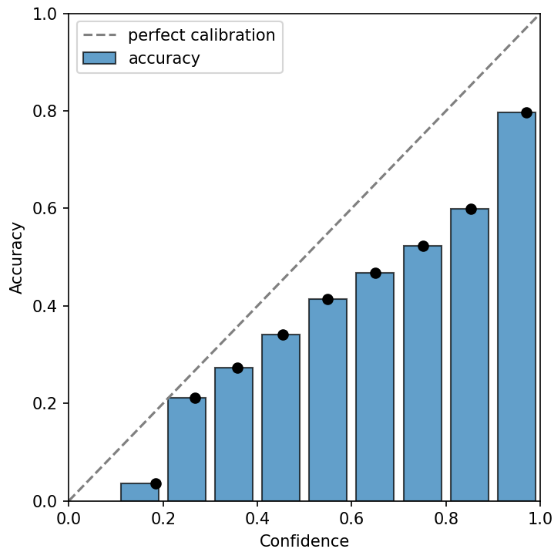
Figure 3: BRACS reliability curves for the balanced reference and severe-condition tail classes on the first locked partition.



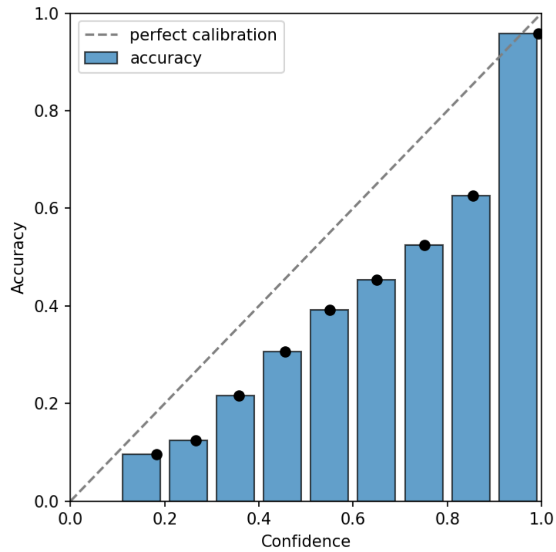
(a) Balanced reference, all classes



(b) Severe, native assignment, tail classes



(c) Severe, aligned assignment, tail classes



(d) Severe, reversed assignment, tail classes

Figure 4: TCGA-UT reliability curves for the balanced reference and severe-condition tail classes on the first locked partition. Both figure sets use ten equal-width confidence bins.

G Do Individual Partitions Change the Conclusion?

Equal-weight pooling is easier to interpret when the individual partitions agree. Tables 27 and 28 show every partition-specific deficit on its own endpoint scale; their averages and paired uncertainty are already reported in Appendix B. The severity checks in Appendix A.2 rule out large allocation-ratio variation as the explanation for these differences.

Dataset	Comparison	Assignment	Partition 1	Partition 2	Partition 3
BRACS	Patient shortage	Native	2.52	0.75	5.47
BRACS	Patient shortage	Aligned	2.52	0.67	5.92
BRACS	Patient shortage	Reversed	2.00	0.47	5.39
BRACS	Moderate	Native	3.67	3.53	2.83
BRACS	Moderate	Aligned	1.63	2.97	-0.12
BRACS	Moderate	Reversed	8.61	6.76	9.14
BRACS	Severe	Native	4.67	5.99	5.69
BRACS	Severe	Aligned	1.19	7.30	4.41
BRACS	Severe	Reversed	12.88	12.87	13.94
BRACS	Severe spread	Native	5.63	9.98	9.68
BRACS	Severe spread	Aligned	5.56	3.84	3.04
BRACS	Severe spread	Reversed	14.29	11.60	15.40
TCGA-UT	Patient shortage	Native	4.14	4.82	5.33
TCGA-UT	Patient shortage	Aligned	4.35	4.69	5.45
TCGA-UT	Patient shortage	Reversed	4.23	4.82	5.46
TCGA-UT	Moderate	Native	0.56	0.36	0.17
TCGA-UT	Moderate	Aligned	0.08	0.13	0.19
TCGA-UT	Moderate	Reversed	0.50	0.73	0.13
TCGA-UT	Severe	Native	1.87	1.66	1.82
TCGA-UT	Severe	Aligned	1.34	1.52	1.14
TCGA-UT	Severe	Reversed	1.40	1.60	1.35
TCGA-UT	Severe spread	Native	5.07	4.55	5.26
TCGA-UT	Severe spread	Aligned	3.05	2.74	3.44
TCGA-UT	Severe spread	Reversed	2.50	2.14	2.58

Table 27: Balanced-accuracy deficit by partition, in percentage points. Positive means damage. Partitions 1–3 correspond to stored partition identifiers 0–2.

Dataset	Comparison	Assignment	Partition 1	Partition 2	Partition 3
BRACS	Patient shortage	Native	0.070	0.171	0.171
BRACS	Patient shortage	Aligned	0.070	0.148	0.202
BRACS	Patient shortage	Reversed	0.060	0.138	0.204
BRACS	Moderate	Native	0.027	0.092	0.437
BRACS	Moderate	Aligned	0.598	0.243	0.298
BRACS	Moderate	Reversed	0.410	0.037	1.814
BRACS	Severe	Native	0.483	0.160	1.420
BRACS	Severe	Aligned	0.306	0.978	0.904
BRACS	Severe	Reversed	1.266	1.567	2.398
BRACS	Severe spread	Native	0.357	0.183	0.075
BRACS	Severe spread	Aligned	1.144	0.703	0.877
BRACS	Severe spread	Reversed	1.489	0.941	0.956
TCGA-UT	Patient shortage	Native	0.209	0.356	0.224
TCGA-UT	Patient shortage	Aligned	0.207	0.343	0.226
TCGA-UT	Patient shortage	Reversed	0.203	0.347	0.224
TCGA-UT	Moderate	Native	-0.197	-0.159	-0.241
TCGA-UT	Moderate	Aligned	-0.125	-0.148	-0.080
TCGA-UT	Moderate	Reversed	-0.069	-0.057	-0.060
TCGA-UT	Severe	Native	-0.190	-0.187	-0.129
TCGA-UT	Severe	Aligned	0.133	0.177	0.119
TCGA-UT	Severe	Reversed	-0.011	-0.091	0.010
TCGA-UT	Severe spread	Native	0.149	0.273	0.156
TCGA-UT	Severe spread	Aligned	0.224	0.137	0.236
TCGA-UT	Severe spread	Reversed	0.063	0.106	0.073

Table 28: Tail-group NLL deficit by partition, in nats. Negative values mean improved probability quality. These partition estimates do not replace pooled patient-bootstrap inference.

BRACS balanced-accuracy damage from patient shortage is smallest in partition 2 and largest in partition 3, so two partitions carry most of the pooled magnitude. TCGA-UT repeats the pooled deficit under a stronger patient-coverage contrast. Agreement on the broad mechanism therefore coexists with substantial uncertainty about its magnitude in a particular split.

H What Additional Computation Was Required?

The useful comparison is additional cost against cross-entropy in the same condition. Tables 29 and 30 summarize each method's minimum and maximum recorded differences across assignments and conditions. Minutes denote accelerator time; MiB denotes peak memory divided by 2^{20}. Examples are processed presentations, Unique is the recorded unique-exposure count, and Parameters is model size. These ranges describe measured variation, not uncertainty intervals; rounded zeros can hide small differences.

Method	Minutes	MiB	Examples	Unique	Parameters
Balanced sampling	0.0	−15.9 to 0.0	0	−4 to 0	0
CE + soft F_1	0.2 to 0.8	−15.9 to 0.0	0	−4 to 0	0
CE + soft MCC	0.1	−15.9 to 15.9	0	−4 to 0	0
CFAL	0.1 to 0.2	1,037.6 to 1,085.3	0	0	−7
Nominal support	0.0	−15.9 to 15.9	0	0	0
cRT	0.1	−19.3 to −3.3	129	0	0
Focal loss	0.0 to 0.1	−15.9 to 15.9	0	0	0
Independent support	0.0	−0.2 to 0.0	0	0	0
Independent support (matched β)	0.0	−0.2 to 0.0	0	0	0
Logit adjustment (train)	0.0	−15.9 to 0.0	0	0	0
OKO	0.0 to 0.1	−16.5 to 31.1	2,727 to 2,983	−80 to 0	3,591
Difficulty	0.0	−15.9 to 15.9	0	0	0
Logit adjustment (post-hoc)	−0.2	−3,331.2 to −3,315.0	−577,400	−19,330	0
Diversity	0.0 to 0.1	−10.9 to 34.1	0	0	0
Prevalence	0.0	−15.9 to 15.9	0	0	0

Table 29: BRACS additional cost against matched cross-entropy. Post-hoc adjustment records incremental work after checkpoint reuse; its negative training counts are not end-to-end savings.

Method	Minutes	MiB	Examples	Unique	Parameters
Balanced sampling	0.0 to 0.1	−15.9 to 0.1	0	−10 to 0	0
CE + soft F_1	1.7 to 3.2	−15.9 to 15.9	0	−10 to 0	0
CE + soft MCC	0.1 to 0.2	−15.9 to 15.9	0	−10 to 0	0
CFAL	0.2 to 0.8	14,514.9 to 14,543.5	0	0	−30
Nominal support	−0.1 to 0.1	−15.9 to 15.9	0	0	0
cRT	0.2 to 0.5	−19.3 to 12.6	43	0	0
Focal loss	0.0 to 0.2	−15.9 to 0.0	0	0	0
Independent support	−0.1 to 0.1	−15.9 to 15.9	0	0	0
Independent support (matched β)	−0.1 to 0.1	−15.9 to 0.1	0	0	0
Logit adjustment (train)	−0.2 to 0.1	−15.9 to 31.8	0	0	0
OKO	0.2 to 0.4	−16.5 to 15.2	396 to 524	−21 to 0	15,390
Difficulty	−0.1 to 0.2	−15.9 to 15.9	0	0	0
Logit adjustment (post-hoc)	−0.7 to −0.5	−18,024.4	−1,340,000	−44,660	0
Diversity	0.0 to 0.2	−14.0 to 37.8	0	0	0
Prevalence	0.0 to 0.1	−15.9 to 0.0	0	0	0

Table 30: TCGA-UT additional cost, with the same units and aggregation as Table 29. Exposure differences and memory overhead vary by mechanism.

Exposure matching removes most differences in data volume, while integral update counts leave residual mismatches for methods such as OKO and two-stage classifier retraining. CFAL has substantially higher peak memory than cross-entropy in both datasets. Post-hoc logit adjustment reuses a trained checkpoint, so its incremental training cost cannot be compared with a complete training run without including that checkpoint's cost. These distinctions are more informative than a single universal ranking of computational expense.

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